

	NEW MAJOR WATER FLOW LINES, AND REVISION CLOUDS
	NEW MINOR WATER FLOW LINES, NEW CHEMICAL FLOW LINES, NEW MAIN EQUIPMENT, NEW CONCRETE BARRIERS
	NEW AIR FLOW LINES, NEW SECONDARY EQUIPMENT,
	NEW VALVES, NEW INSTRUMENTATION SYMBOLS, NEW FITTINGS, NEW INSTRUMENTATION LINES
	FUTURE FLOW LINES & EQUIPMENT
	EXISTING FLOW LINES & EQUIPMENT
	PACKAGE UNIT/SKID LIMITS
	PNEUMATIC LINES
	DATALINK
	DCS-PLC
	HYDRAULIC WATER LINES
	CAPILLARY
	EQUIPMENT DETAILS (HATCHING, INTERIOR COMPONENTS, ETC.) BY CUSTOMER

LINE CONTINUATIONS

LINE CONTINUATION

	-	BLOWER OR AIR HANDLING	MD	-	MEMBRANE DEGASIFIER CONTACTOR
CDI	-	CONTINUOUS DEIONIZATION CELL PACK	MMF	-	MULTI-MEDIA FILTER
CFM	-	CUBIC FEET PER MINUTE	MOC	-	MATERIAL OF CONSTRUCTION
CS	-	CARBON STEEL	MW	-	MANWAY
CW	-	CITY WATER (POTABLE)	NO	-	NORMALLY OPEN
DA	-	STEAM DEAERATOR	NC	-	NORMALLY CLOSED
DIA	-	DIAMETER	NF	-	NANOFILTER
DWG	-	DRAWING	OD	-	OUTSIDE DIAMETER
EPSS	-	ELECTRO POLISHED STAINLESS STEEL	OZ	-	OZONE GENERATOR
F	-	FILTER VESSEL (CARTRIDGE, MEDIA, CARBON, PRECOAT, AIR)	P	-	PUMP
FC	-	FAIL CLOSED	PE	-	POLYETHYLENE
FDD	-	FORCED DRAFT DECARBONATOR VESSEL	PP	-	POLYPROPYLENE
FO	-	FAIL OPEN	PSIA	-	POUNDS PER SQUARE INCH - ABSOLUTE
FRP	-	FIBERGLASS REINFORCED PLASTIC	PSID	-	POUNDS PER SQUARE INCH - DIFFERENTIAL
GAL	-	GALLONS	PSIG	-	POUNDS PER SQUARE INCH - GAUGE
GPD	-	GALLONS PER DAY	PVC	-	POLY VINYL CHLORIDE
GPH	-	GALLONS PER HOUR	PVDF	-	POLYVINYLIDENE FLUORIDE
GPM	-	GALLONS PER MINUTE	PW	-	PLANT WATER
HG	-	INCHES OF MERCURY	RO	-	REVERSE OSMOSIS
HOA	-	HAND/OFF/AUTO	RPM	-	REVOLUTIONS PER MINUTE
HP	-	HORSEPOWER	SB	-	SPRAYBALL
HDPE	-	HIGH DENSITY XXXX XXXX	SCFM	-	STANDARD CUBIC FEET PER MINUTE
HX	-	HEAT EXCHANGER	SCH	-	SCHEDULE
IA	-	INSTRUMENT AIR	SG	-	SPECIFIC GRAVITY
ID	-	INSIDE DIAMETER	SOV	-	SOLENOID VALVE
IX	-	ION EXCHANGER	SP	-	SETPOINT
M	-	MIXER (AGITATOR, AERATOR)	SS	-	STAINLESS STEEL

	-	MEMBRANE DEGASIFIER CONTACTOR	X -	Y - DIAPHRAGM
HMF	-	MULTI-MEDIA FILTER		
IIOC	-	MATERIAL OF CONSTRUCTION		
IW	-	MANWAY		
IO	-	NORMALLY OPEN		
IC	-	NORMALLY CLOSED		
IF	-	NANOFILTER		
ID	-	OUTSIDE DIAMETER		
OZ	-	OZONE GENERATOR		
P	-	PUMP		
PE	-	POLYETHYLENE		
PP	-	POLYPROPYLENE		
PSIA	-	POUNDS PER SQUARE INCH - ABSOLUTE		
PSID	-	POUNDS PER SQUARE INCH - DIFFERENTIAL		
PSIG	-	POUNDS PER SQUARE INCH - GAUGE	SSH -	STRAIGHT SIDE HEIGHT
PVC	-	POLY VINYL CHLORIDE	SST -	STAINLESS STEEL TUBE
PVDF	-	POLYVINYLIDENE FLUORIDE	SV -	SAMPLE VALVE
PW	-	PLANT WATER	SW -	SEAL WATER
RO	-	REVERSE OSMOSIS	T -	TANK (LIQUID OR CHEMICAL STORAGE; NEUTRALIZATION)
RPM	-	REVOLUTIONS PER MINUTE	TDH -	TOTAL DYNAMIC HEAD (FEET OF FLUID)
SB	-	SPRAYBALL	TYP -	TYPICAL
SCFM	-	STANDARD CUBIC FEET PER MINUTE	UF -	ULTRAFILTER MODULE
SCH	-	SCHEDULE	UV -	ULTRAVIOLET LIGHT (STERILIZER, TOC OR O ₃ REMOVAL)
SG	-	SPECIFIC GRAVITY	VAC -	VACUUM
SOV	-	SOLENOID VALVE	VD -	VACUUM DEGASIFIER VESSEL
SP	-	SETPOINT	VFD -	VARIABLE FREQUENCY DRIVE
SS	-	STAINLESS STEEL	WC -	WATER COLUMN

	-	CYLINDER
	-	DIAPHRAGM
	-	DOUBLE DIAPHRAGM
	-	ELECTRO-PNEUMATIC
	-	MOTOR
	-	POSITIONER
	-	RELIEF
	-	SOLENOID
	-	MANUAL ACTUATOR
	-	FAIL CLOSED
	-	FAIL OPEN
	-	LIMIT STOP

	-	3-WAY
	-	4-WAY
	-	AIR RELEASE
	-	AIR RELIEF
	-	BALL
	-	BUTTERFLY
	-	BLOCK & BLEED
	-	CHECK
	-	DIAPHRAGM
	-	GATE
	-	GLOBE
	-	KNIFE
	-	NEEDLE
	-	PLUG
	-	PRESSURE REDUCING
	-	RELIEF
	-	VACUUM BREAKER (STYLE 1)
	-	CONSERVATION
	-	Y - DIAPHRAGM

	-	DIAPHRAGM SEAL
	-	DRAIN
	-	EDUCTOR/EJECTOR
ED		
	-	EXPANSION JOINT
EXP		
	-	FLANGE
	-	FLEXIBLE HOSE
H		
	-	HOSE CONNECTION
C		
	-	INSULATION
	-	INSULATED PIPE WITH ELECTRIC HEAT TRACE
F		
	-	INSULATED PIPE WITH STEAM HEAT TRACE
S		
	-	ORIFICE
	-	UNION
	-	PIGTAIL
	-	PULSATION DAMPENER
R		
	-	PUMP
	-	METERING PUMP
	-	QUICK DISCONNECT
	-	REDUCER (CONCENTRIC)
	-	REDUCER (ECCENTRIC)
	-	RUPTURE DISK
	-	SANITARY COUPLING
	-	SIGHT FLOW INDICATOR
X		
	-	SIGHT FLOW STRAINER
S		
	-	SPRAYBALL
	-	STATIC MIXER
M		
	-	STEAM TRAP
ST		
	-	STRAINER
S		
	-	VIC COUPLING

	-	MAGNETIC FLOW METER
	-	SONIC FLOW METER
	-	TURBINE FLOW METER
	-	VORTEX FLOW METER
	-	PADDELEWHEEL FLOW METER
	-	ROTAMETER
	-	PILOT LIGHT
	-	VARIABLE FREQUENCY DRIVE
	-	CALIBRATION COLUMN
	-	ULTRASONIC SENSOR

	PRIMARY LOCATION NORMALLY ACCESSIBLE TO OPERATOR (IE: LOCAL PANEL MOUNTED)	FIELD MOUNTED (IE: PIPE MOUNTED)	AUXILIARY LOCATION NORMALLY ACCESSIBLE TO OPERATOR (IE: REMOTE PANEL MOUNTED)	LOCATION NORMALLY INACCESSIBLE TO OPERATOR (IE: MOUNTED INSIDE PANEL)	BY OTHERS**
DISCRETE INSTRUMENTS	1 	2 	3 	4 	5
SHARED DISPLAY, SHARED CONTROL	6 	7 	8 	9 	10
COMPUTER FUNCTION	11 	12 	13 	14 	15
PROGRAMMABLE LOGIC CONTROL	16 	17 	18 	19 	20

* ABBREVIATIONS OF THE USER'S CHOICE SUCH AS IP1 (INSTRUMENT PANEL #1), IC2 (INSTRUMENT CONSOLE #2), CC3 (COMPUTER CONSOLE #3), ETC., MAY BE USED WHEN IT IS NECESSARY TO SPECIFY, INSTRUMENT OR FUNCTION LOCATION.

** PRIMARY LOCATION SYMBOL FOR EACH INSTRUMENT TYPE IS SHOWN AS AN EXAMPLE OF "BY OTHERS"

*** AI,AO,DI,DO

TIC-SCH10-01- INSTRUMENT IDENTIFICATION OR TAG NUMBER
 LOOP NUMBER
 MATERIAL IDENTIFIER
 FUNCTIONAL IDENTIFICATION

WHERE:
(LEGEND BASED ON ISA STANDARD S 5.1)

ISA INSTRUMENT IDENTIFICATION TABLE			
	FIRST LETTER		SUCCEEDING LETTERS
	PROCESS VARIABLE	MODIFIER (IF NEEDED)	READOUT OR COMPUTER FUNCTION
A	ANALYSIS		ALARM
B	BURNER, COMBUSTION		USER'S CHOICE
C	USER'S CHOICE		CONTROL
D	USER'S CHOICE	DIFFERENTIAL	
E	VOLTAGE		SENSOR (PRIMARY ELEMENT)
F	FLOW RATE	RATIO (FRACTION)	
G	USER'S CHOICE		GLASS, VIEWING DEVICE
H	HAND		HIGH
I	CURRENT (ELECTRICAL)		INDICATE
J	POWER	SCAN	
K	TIME, TIME SCHEDULE	TIME RATE OF CHANGE	CONTROL STATION
L	LEVEL		LIGHT
M	USER'S CHOICE	MOMENTARY	LOW
N	USER'S CHOICE		MIDDLE, INTERMEDIATE
O	USER'S CHOICE		USER'S CHOICE
P	PRESSURE, VACUUM		ORIFICE (RESTRICTION)
Q	QUANTITY	INTEGRATE, TOTALIZE	POINT (TEST CONNECTION)
R	RADIATION		
S	SPEED, FREQUENCY	SAFETY	RECORD
T	TEMPERATURE		SWITCH
U	MULTIVARIABLE		TRANSMIT
V	VIBRATION, MECH. ANALYSIS		MULTIFUNCTION
W	WEIGHT, FORCE		MULTIFUNCTION
X	UNCLASSIFIED	X-AXIS	VALVE, DAMPER, LOUVER
Y	EVENT, STATE OR PRESENCE	Y-AXIS	WELL
Z	POSITION, DIMENSION	Z-AXIS	UNCLASSIFIED
			UNCLASSIFIED
			RELAY, COMPUTE, CONVERT
			DRIVER, ACTUATOR, UNCLASSIFIED FINAL CONTROL ELEMENT

PRELIMINARY^A
NOT FOR CONSTRUCTION

STD: BORDER-0106-24X36D1

	INTL REF:
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 BAR = 1" AT PLOT SCALE

A	INITIAL RELEASE FOR PROPOSAL	06/03/16	MLS			
REV	DESCRIPTION	DATE	DWN	CHKD	APVD	ECN

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DESIGNER	DATE
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MANAGER	DATE
FILE:	
SCALE:	NONE

TITLE	PROCESS AND INSTRUMENTATION DIAGRAM LEGEND SHEET ONE
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	CLIENT



WATER TECHNOLOGIES
COLORADO SPRINGS, CO
719-570-9600

	PRO
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SUBJECT	
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CODE

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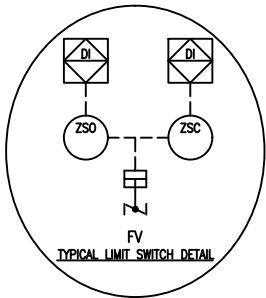
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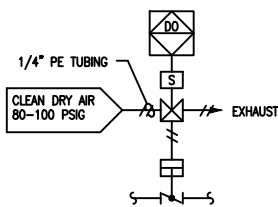
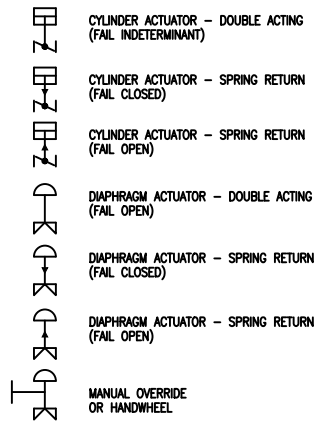
8	REV A
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PUMP DIAGRAM NOTES:
1. HAND-OFF-AUTO (H-O-A) COULD BE REPLACED BY HAND-OFF (H-O).
2. MOTOR CONTROL CENTER (MCC), MOTOR STARTER (MS), VARIABLE FREQUENCY DRIVE (VFD) OR DISCONNECT SWITCH (DS) COULD BE BY CUSTOMER OR EVOQUA WATER TECHNOLOGIES.
3. ZS INDICATES VOLUME ADJUSTMENT, EITHER FREQUENCY OR STROKE LENGTH. EXAMPLE INDICATES ADJUSTMENT FROM THE PLC.
4. THE POSITIVE DISPLACEMENT PUMP CAN HAVE THE SAME COMBINATIONS AS SHOWN FOR THE CENTRIFUGAL PUMP.

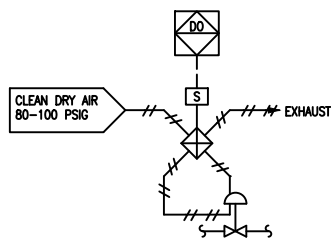


ZSO ← AS SHOWN IN P & ID
ZSC

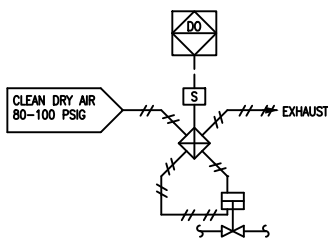
AUTO VALVE ACTUATORS



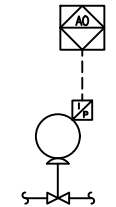
TYPICAL AIR ACTUATED VALVE CONTROL DETAIL (AIR SUPPLY NOT SHOWN ON SYSTEM P&ID)



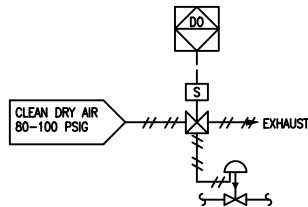
TYPICAL VALVE SCHEMATIC FOR ALL VALVES W/AIR TO DIAPHRAGM ACTUATOR



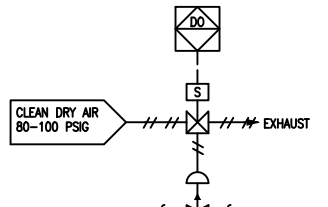
TYPICAL VALVE SCHEMATIC FOR ALL VALVES W/AIR TO AIR CYLINDER ACTUATOR



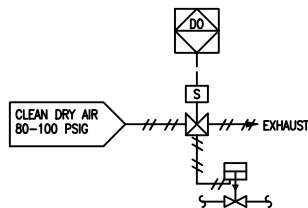
TYPICAL PROPORTIONAL CONTROL VALVE



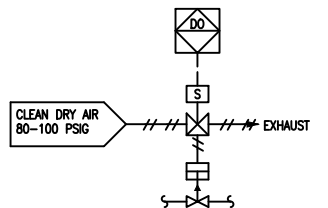
TYPICAL VALVE SCHEMATIC FOR ALL VALVES W/SPRING TO CLOSE DIAPHRAGM ACTUATOR



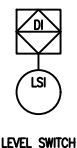
TYPICAL VALVE SCHEMATIC FOR ALL VALVES W/SPRING TO OPEN DIAPHRAGM ACTUATOR



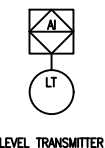
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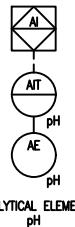
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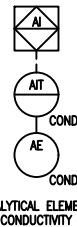
LEVEL SWITCH



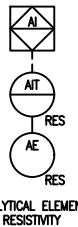
LEVEL TRANSMITTER



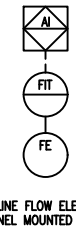
ANALYTICAL ELEMENT pH



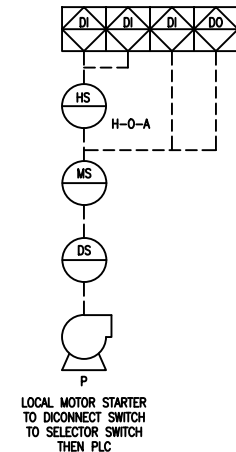
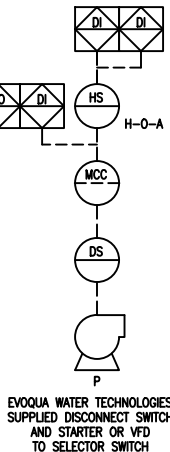
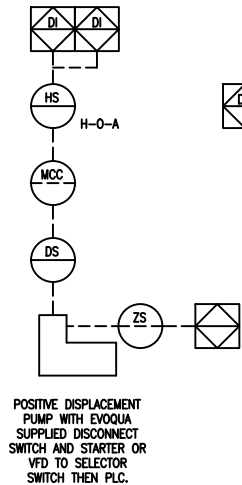
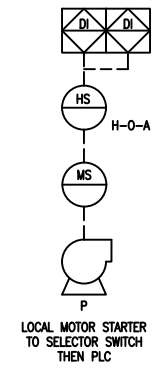
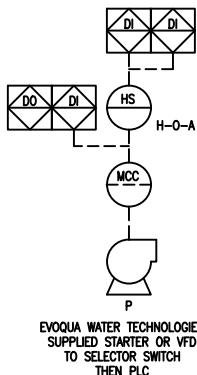
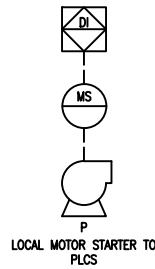
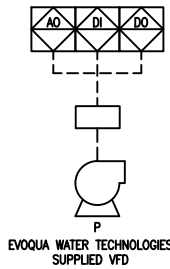
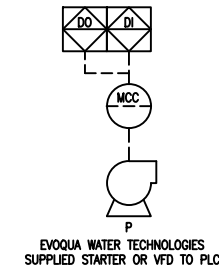
ANALYTICAL ELEMENT CONDUCTIVITY



ANALYTICAL ELEMENT RESISTIVITY



INLINE FLOW ELEMENT PANEL MOUNTED FLOW METER W/ANALOG INPUT

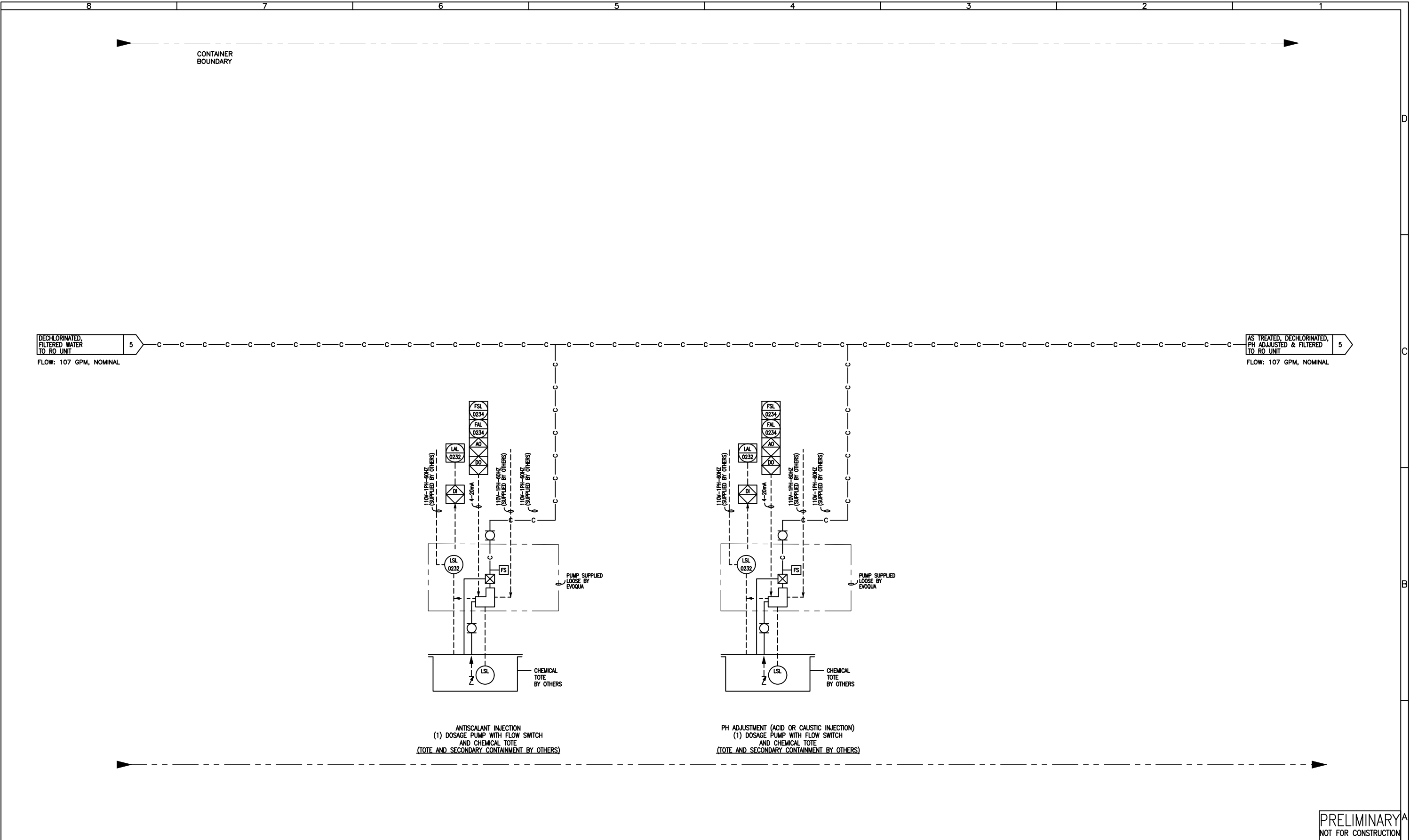


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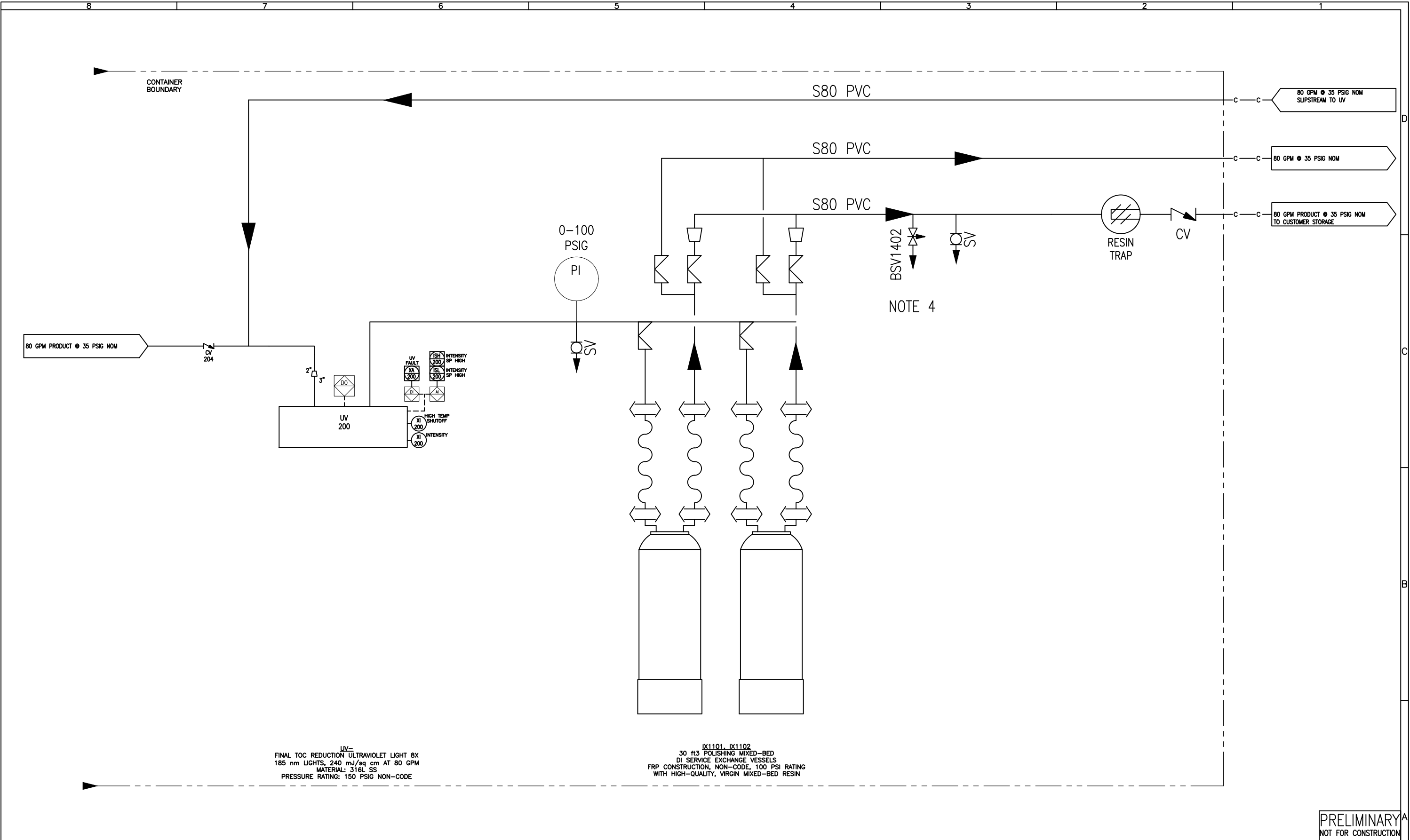
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DESIGNER DATE
CHECKER DATE
ENGINEER DATE
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FILE:
SCALE: NONE

TITLE PROCESS AND INSTRUMENTATION DIAGRAM
LEGEND SHEET TWO
CLIENT
PROJECT CODE DRAWING SHEET 2 OF 8 REV A
EVOQUA WATER TECHNOLOGIES
COLORADO SPRINGS, CO
719-570-9600



STD: BORDER-0106-24X36D1		INTL REF:		BAR = 1" AT PLOT SCALE		- SEE SHEET ONE		REV		DESCRIPTION		DATE		DWN		CHKD		APVD		ECN		COMPANY CONFIDENTIAL THIS DOCUMENT AND ALL INFORMATION CONTAINED HEREIN ARE THE PROPERTY OF EVOQUA AND/OR ITS AFFILIATES. THE DESIGN CONCEPTS AND INFORMATION CONTAINED HEREIN ARE PROPRIETARY TO EVOQUA AND ARE SUBMITTED IN CONFIDENCE. THEY ARE NOT TRANSFERABLE AND MUST BE USED ONLY FOR THE PURPOSE FOR WHICH THE DOCUMENT IS EXPRESSLY REPRODUCED, LOANED OR USED IN ANY OTHER MANNER WITHOUT THE EXPRESS WRITTEN CONSENT OF EVOQUA. IN NO EVENT SHALL THEY BE USED IN ANY MANNER DETRIMENTAL TO THE INTEREST OF EVOQUA. ALL PATENT RIGHTS ARE RESERVED. UPON THE DEMAND OF EVOQUA, THIS DOCUMENT, ALONG WITH ALL COPIES AND EXTRACTS, AND ALL RELATED NOTES AND ANALYSES, MUST BE RETURNED TO EVOQUA OR DESTROYED, AS INSTRUCTED BY EVOQUA. ACCEPTANCE OF THE DELIVERY OF THIS DOCUMENT CONSTITUTES AGREEMENT TO THESE TERMS.		DESIGNER DATE		CHECKER DATE		ENGINEER DATE		MANAGER DATE		FILE:		SCALE: NONE		TITLE PROCESS AND INSTRUMENTATION DIAGRAM		CLIENT		PROJECT		CODE		DRAWING		SHEET 4 OF 8		REV A	
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UV=
FINAL TOC REDUCTION ULTRAVIOLET LIGHT 8X
185 nm LIGHTS, 240 mJ/sq cm AT 80 GPM
MATERIAL: 316L SS
PRESSURE RATING: 150 PSIG NON-CODE

IX1101, IX1102
30 ft3 POLISHING MIXED-BED
DI SERVICE EXCHANGE VESSELS
FRP CONSTRUCTION, NON-CODE, 100 PSI RATING
WITH HIGH-QUALITY, VIRGIN MIXED-BED RESIN

PRELIMINARY
NOT FOR CONSTRUCTION

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